

To,

The Chief Engineer (Legal)
Legal Division, Central Electricity Authority
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Subject: Objections and suggestions on the draft Central Electricity Authority (Technical Standards for Construction of Electric Plants and Electric Lines) 2nd Amendment, Regulations, 2026 — F. No. CEA-TH-17/1/2021-TETD Division, dated 3 September 2026

Sir / Madam,

I publish independent, primary-source research on renewable open-access and banking entitlements in India, at dsnikam.in. I have no commercial interest in any generator, developer, supplier, licensee or financier, and I accept no fee, commission or consideration from any party in this field. These comments are submitted in my personal capacity under sub-section (3) of section 177 of the Electricity Act, 2003.

I support the direction of the draft. I offer three comments: one interaction the draft does not address, one clarification within the Authority's remit, and one point of drafting.

1. In at least one State, the mandated storage will be commissioned into that State's harshest deviation regime

The first proviso to the proposed Regulation 106B(20) requires a ground-mounted solar or on-shore wind plant commissioned after 1 July 2027 to be equipped with a co-located Energy Storage System of at least two hours' duration and a capacity of at least ten per cent of the installed capacity of the plant, rising to four hours under the second proviso for plants commissioned after 1 July 2029 and up to 30 June 2031.

Karnataka has notified the *Karnataka Electricity Regulatory Commission (Forecasting, Scheduling, Deviation Settlement Mechanism and related matters for Sellers of Wind, Solar and WS-Hybrid Generation sources) Regulations, 2026* — Notification No. KER/WS(DSM)/2026-27/2150 dated 19 June 2026, published in the Karnataka Gazette of 22 June 2026 (Vol. 161, Issue 120), replacing the 2015 regulations. Clause 13(e) of those regulations sets the permissible deviation as follows:

Generator / seller	Deviation range
Wind generators / sellers	± 10% (within the limits of >90% and <110%)
Solar generators / sellers	± 5% (within the limits of >95% and <105%)
Hybrid generators / sellers	± 10% (within the limits of >90% and <110%)
ESS (with wind or solar or WS-hybrid) generators / sellers	Nil

The clause provides that no DSM charges arise if deviation is within those limits.

The consequence is that **the asset the Authority proposes to make compulsory is, in that State, the only class of asset with no tolerance band at all**. A wind plant may deviate by ten per cent without charge and a solar plant by five; a storage system attached to either has none.

By clause 4 of the same regulations, they apply to sellers whether supplying the distribution licensees, or supplying third parties under open access, or for captive use, within the State and outside it — for wind at

10 MW and above, and for solar and wind-solar hybrid at 5 MW and above. The position therefore reaches the projects a co-located storage requirement will most directly affect.

Clause 4 also carries a note that in the case of an ESS, the battery capacity shall not exceed the MW connectivity granted at the interface point. **That limit is expressed in MW; the Authority's requirement is expressed in hours of storage at a percentage of installed capacity.** The two instruments measure the same asset on different axes. At two hours this is unlikely to bind. At four hours, for a plant whose interface capacity is already fully subscribed, it may.

I am not suggesting the Authority should regulate State deviation frameworks, and I recognise that it cannot. I am submitting that a technical standard which makes an asset compulsory should issue with the operating environment of that asset on the record.

Suggestion. That the explanatory material accompanying the final regulations record the interaction between a mandated co-located ESS and State deviation, scheduling and banking frameworks; and that the Authority place the point before the Forum of Regulators, so that the deviation treatment of storage installed in compliance with a central mandate is considered rather than arrived at State by State after commissioning.

2. A clarification within the Authority's remit

The draft does not state at what point energy **discharged from the mandated co-located storage** is deemed generated for scheduling, metering and accounting — at the hour of discharge, or at the hour of charging and delivered later.

This is not a drafting nicety. Where banked energy is settled slot by slot, the answer decides what the energy is worth. Clause 12.e(ii) of the *Karnataka Electricity Regulatory Commission (Terms and Conditions for Open Access) Regulations, 2025* provides:

“The energy banked during TOD Slot-1 & Slot-3 (Time period of the ToD Slots shall be as defined in the Tariff Orders issued by the Commission from time to time) shall be permitted to be drawn during TOD Slot-1, Slot-2, Slot-3 and Slot-4. The energy banked during TOD Slot-2 & Slot-4 shall be permitted to be drawn during TOD Slot-2 & Slot-4 only.”

The illustration at clause 12.e(iv) sets out the slot periods: Slot-1 is 6 am to 9 am, Slot-2 is 9 am to 6 pm, Slot-3 is 6 pm to 10 pm and Slot-4 is 10 pm to 6 am. Slot-2 is therefore the solar period and Slot-3 the evening peak. Energy banked in the solar hours cannot be drawn in the evening peak at all, and the regulations contain no provision permitting it on payment of any additional charge. Whether a co-located battery alters that turns entirely on whether its discharge is treated as occurring at the hour of discharge or the hour of charging — and no instrument I have read answers it.

If the Authority does not settle this, it will be settled differently in different States, and discovered by developers after commissioning rather than before investment.

Suggestion. That the final regulations, or the connectivity and metering standards read with them, state expressly at what point energy from a mandated co-located ESS is deemed generated for scheduling and accounting, and specify the metering arrangement required to establish it.

3. The four-hour requirement lapses on 30 June 2031

The second proviso applies the four-hour requirement to plants commissioned **after 1 July 2029 and up to 30 June 2031**. The first proviso, requiring two hours, carries no end date.

Read together, a plant commissioned on 1 July 2031 appears to fall back to the two-hour requirement, having been preceded by plants required to carry four. If that is intended, the reason is not apparent from the draft; if it is not, the second proviso needs an open end or a stated successor.

A developer taking a financing decision in 2027 for a plant commissioning in 2031 or later cannot size the asset on the draft as it stands, and the uncertainty falls on the part of the pipeline with the longest lead time.

Suggestion. That the position for plants commissioned on or after 1 July 2031 be stated expressly, even if only as an intention to review before that date.

A related drafting point. Clause 2 of the draft reads that after regulation 106B(19), “the following regulation substituted”. As 106B(20) is being added rather than replaced, “inserted” appears to be intended.

Basis of these comments

The two Karnataka regulations cited above were read from the notified texts. The wider State-level position — banking charges, drawal restrictions, quantum caps and expiry in Karnataka, Gujarat and Andhra Pradesh — is set out with its sources in *India Renewable Banking Index*, Report No. 2, published on 13 September 2026 at dsnikam.in/india-renewable-banking-index-2, which is free and open to correction.

I would be glad to provide the underlying references if that is of use to the Authority.

Yours faithfully,

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